

PNL Assignment on Byte pair encoding

BPE on the corpus:

5 nation, 3 national, 2 nationality, 1 international

Initial representation

5 nation_

3 national_

2 nationality_

1 international_

Iteration 1

Pair counts (word frequency × occurrences in the word):

Pair	Calculation	Freq
n a	$5 + (3 \times 2) + (2 \times 2) + (1 \times 2)$	17
a t	$5 + 3 + 2 + 1$	11
t i	$5 + 3 + 2 + 1$	11
i o	$5 + 3 + 2 + 1$	11
o n	$5 + 3 + 2 + 1$	11
a l	$3 + 2 + 1$	6
n _	5	5
l _	$3 + 1$	4
l i, i t, t y, y _	2 each	2
i n, n t, t e, e r, r n	1 each	1

Merge: n a → na (17)

Now the words become:

5 nation_

3 nationnal_

2 nationnality_

1 internationnal_

Iteration 2

Pair	Calculation	Freq
na t	$5 + 3 + 2 + 1$	11
t i	$5 + 3 + 2 + 1$	11
l o	$5 + 3 + 2 + 1$	11
o n	$5 + 3 + 2 + 1$	11
n na	$3 + 2 + 1$	6
na l	$3 + 2 + 1$	6
n _	5	5
l _	$3 + 1$	4
ll, lt, ty, y _	2 each	2
l n, n t, t e, e r, r na	1 each	1

Four pairs tie at 11 (na t, t i, i o, o n). The tie-break picks the one met first, reading word 1 left to right, which is na t.

Merge: na t → nat (11)

Now the words become:

5 nation _

3 nationnal _

2 nationality _

1 international _

Iteration 3

Pair	Calculation	Freq
nat i	$5 + 3 + 2 + 1$	11
i o	$5 + 3 + 2 + 1$	11
o n	$5 + 3 + 2 + 1$	11
n na	$3 + 2 + 1$	6
na l	$3 + 2 + 1$	6
n _	5	5
l _	$3 + 1$	4
li, it, ty, y _	2 each	2
i n, n t, t e, e r, r nat	1 each	1

nat i, i o and o n tie at 11. In word 1, nat i comes first.

Merge: nat i $\rightarrow$ nati (11)

Now the words become:

5 nati o n _

3 nati o n n a l _

2 nati o n n a l i t y _

1 i n t e r n a t i o n n a l _

Summary table

Iteration	Pair merged	Pair frequency	New token
1	n + a	17	na
2	na + t	11 (tie-break)	nat
3	nat + i	11 (tie-break)	nati

Final corpus after 3 merges:

5 nati o n _

3 nati o n n a l _

2 nati o n n a l i t y _

1 i n t e r n a t i o n n a l _

The counts show that n a is the most frequent pair (17), since "na" appears twice in most of the longer words. After that, several pairs tie at 11, so the tie-breaking rule decides the merges.